

R.O.B. VISION

VIRTUAL ROBOT COMPANION / MK.01



USER GUIDE

Play, preview, setup, and problem solving

VERSION 0.1.2

UNO Q · RETROPIE · BATOCERA

CHAPTER 01 / STORY

Meet Buddy

Buddy lived in the dark between game frames. A flash told him to turn; another told him to lift. He could move the pieces, but the edge of the screen was as far as he had ever gone.



Then a little grid of lights appeared beyond the game. Buddy blinked back. The lights became eyes on the UNO Q, and a larger window opened. For the first time, he could see his hands, the game table, and someone waiting for him to move.

Now each signal gives Buddy another step into the room. When the game ends, he waits for the next round.

CHAPTER 02 / USER GUIDE

R.O.B. Vision user guide

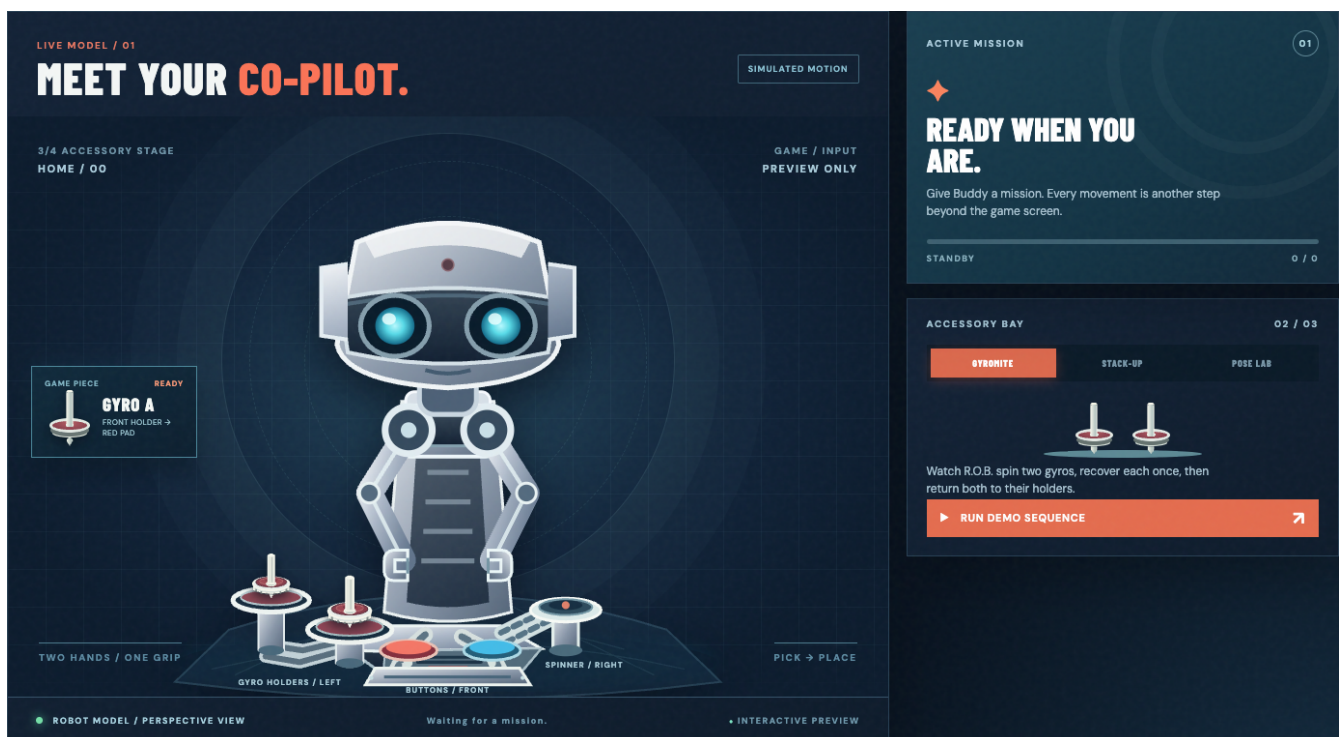
The UNO Q controls Buddy, the virtual robot shown in Mission. RetroPie or Batocera sends Gyromite and Stack-Up commands through the game-frame link. The robot, gyros, blocks, trays, and spinner are graphics and software state. Start with the [Buddy story](#) if you want to meet him, or follow these instructions to set up and play.

Open R.O.B. Vision

Install version 0.1.2 with the two commands in the [installation guide](#): one on the UNO Q, one on the console. The console command prints first-time pairing details for Setup and configures its receiver automatically.

Start **R.O.B. Vision** in UNO Q App Lab, then open `http://arduinoain.local/dashboard/` on a laptop, phone, or tablet. Stop VirtualGlove first if it is running; this UNO Q runs one App Lab app at a time. The direct `http://arduinoain.local:8766/dashboard/` address reaches the same controller. A `file://` page is an offline preview.

Mission shows R.O.B., Accessory Bay, Game Table, Pose Preview, System Vitals, and Activity Feed. **GYROMITE / CONNECTED** means the browser reaches the UNO Q with Gyromite selected. **RETROPIE ONLINE** means the receiver has contacted the UNO Q recently. **GAME FRAMES LINKED** means the selected game is sending its rendered frames.



Try a demonstration

When no supported game is running, select **Gyromite** or **Stack-Up** and **Run Demo Sequence**. Gyromite demonstrates a held unspun gyro, two spinning gyros on pads, spin-down, re-spin, and return to holders. The illustrative spin clock is 55 seconds. Stack-Up moves red alone and then a blue/white group. **Home** resets the pieces. Pose Lab is a character-only movement sandbox. A live game launch stops the demo and selects that game's scene.

Choose an NES emulator

For automatic R.O.B. movement, select **lr-robvision-fceumm** or **lr-robvision-nestopia** for the game in RetroPie. Each entry uses the corresponding installed original core and forwards the game's rendered light cells to the UNO Q. The plain **lr-fceumm** and **lr-nestopia** entries play normally without the R.O.B. Vision frame link. Gyromite and Stack-Up currently default to the FCEUmm choice on the test RetroPie. See the [RetroPie setup guide](#).

Play Gyromite

Open [Setup](#) to check RetroPie and game-frame status. Start Gyromite and watch Mission for **GAME FRAMES LINKED**. R.O.B. follows complete commands from the game. A spinning virtual gyro on a pad keeps its gate pressed while R.O.B. moves elsewhere; an unspun gyro can press one gate while held there. The UNO Q sends those pad states to RetroPie's virtual Controller 2.

Fast Gates responds immediately while Gyromite is selected. Tap **Lower Blue** or **Lower Red** to press, tap again to release, or choose **Release Both**. Keyboard shortcuts are **2**, **1**, and **0**. Holds expire after 60 seconds; both buttons release on game exit or lost receiver data. If regular controls moved a gyro, use **Home** to restore its holder before Fast Gates. Both gate colors have been verified in Game A under FCEUmm. The game screen remains the authority for Hector's position and gate animation.

Play Stack-Up

The live model begins with five colored blocks on Tray 3. In Direct mode, jumping Hector onto a command tile sends **Left**, **Right**, **Up**, **Down**, **Open**, or **Close** through the frame link. Manual controls use the same virtual model. A lower grip carries the contacted block and all blocks above it. Impossible moves are rejected and leave pieces in place. **Home** restores the initial stack. The model does not score the game or initialize Bingo's distinct historical layout. Stack-Up has no Gyromite-style Controller 2 gate return.

Test mode and reset

Launch the game through its R.O.B. Vision emulator choice and enter Test mode. The game-frame link makes R.O.B.'s red light blink automatically on Mission and Setup when it recognizes the Test signal. A separate ready-light command makes the light steady briefly. Exiting Test mode returns the UNO Q matrix to the game's eyes. Neither signal moves R.O.B. **Preview Red Light** on Setup demonstrates the animation without claiming a game signal.

Home resets the selected virtual game. Live **Emergency Stop** clears the game selection; in the offline preview it cancels the script until reset. A browser reload restores the UNO Q's current snapshot. See [Setup](#), [gameplay](#), and [troubleshooting](#).

CHAPTER 03 / USER GUIDE

R.O.B. Vision gameplay guide

Status: scripted preview and connected UNO Q controller. R.O.B., the gyros, spinner, pads, trays, and blocks exist only in the virtual model. The UNO Q controller owns their state and updates the browser. Red and blue pad presses reached RetroPie's virtual joystick, and the player confirmed both gate colors respond in Game A. The RetroPie frame link decoded all six command types in both games during live Direct play.

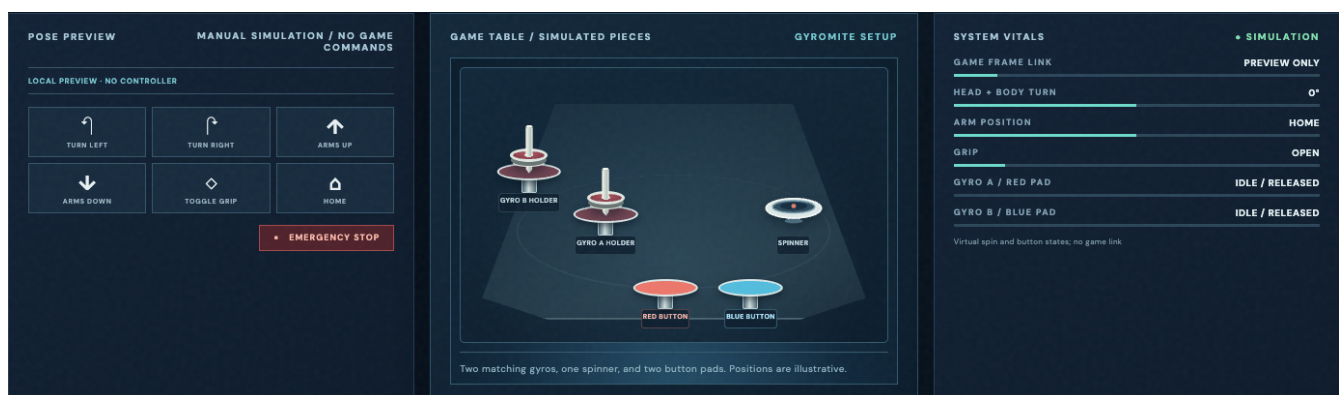
Gyromite

The original Gyromite booklet describes **Test**, **Direct**, **Game A**, and **Game B**. Direct sends six R.O.B. actions: up, down, left, right, open, and close. In Game A, Start switches into the robot-transmission screen for a command and then back to the professor. In Game B, the professor walks while Controller 1 directly commands R.O.B. A gyro on the blue or red tray lowers the matching gate.

The manual explicitly says a gyro **does not need to spin to operate only one gate**. R.O.B. can hold an unspun gyro on the virtual pad, then lift it to release that pad. If both gates must be controlled, spinning one gyro lets it stay upright on one pad while R.O.B. uses his hands to put the other on the other pad. A gyro that is no longer needed belongs on its holder. The spinner is a means to free R.O.B.'s hands, not an obligatory step for every pad press.

Available preview: the finite Gyromite demo shows an unspun held press, a two-gyro spin relay, separate spin-down and one recovery for each gyro, then a tidy return to both holders. It stops at COMPLETE. The spin lifetime is illustrative. Select **Home** to reset. In the connected setup, UNO Q pad state reaches RetroPie's virtual Controller 2; the browser's local-only preview does not drive that receiver.

Live controller: the UNO Q maintains the locations and spin phases of Gyro A and B. It derives red and blue virtual button states from those modeled conditions and sends them over LAN to the paired RetroPie Controller 2 receiver. It releases both buttons on game exit, connection loss, stale packets, or mode change. The player confirmed that both red and blue gate controls work in Game A with the corrected Controller 2 mapping. The **Fast Gates** dashboard controls place or return the matching virtual gyro and send the pad state immediately, with a 60-second automatic release. This keeps gate operation responsive while the player moves Hector. The RetroPie frame link controls R.O.B.'s virtual actions from complete rendered-frame commands; the launch hook supplies the game ID. The controller keeps game context and modeled piece state distinct from unknown on-screen professor or gate positions.



Pose Preview, Game Table, and System Vitals in the local Gyromite preview. Fast Gates appear during live Gyromite play.

Stack-Up

The [Stack-Up booklet](#) describes Direct, Memory, and Bingo. The virtual fixture has five numbered trays and five colored blocks. Direct and Memory start with all five on Tray 3, top to bottom **red, white, blue, yellow, green**. One-player Bingo historically uses any color order on Tray 3; two-player Bingo starts with three on Tray 3 and one each on Trays 2 and 4. The current live model initializes the Direct/Memory stack for all modes; Bingo-specific layouts are not implemented. Direct moves Professor Hector onto six command keys: **left, right, up, down, open, close**. Memory plays a programmed command series; Bingo sends a command when one row or column completes. The UNO Q code applies each validated optical command as one virtual action; all six command types were observed through the live RetroPie frame link. It must model the grippers' height, so R.O.B. can carry a block together with all blocks above it. Stack-Up's documented modes do not need Gyromite's virtual Controller 2 pad return path, even though two-player Bingo uses a second human controller.

Available preview: R.O.B. first moves red alone from Tray 3 to Tray 4, then grips blue and white together and carries both to Tray 2. Green and yellow remain on Tray 3. All five blocks appear in the Live Model, Accessory Bay, and Game Table as they move. The **Stack-Up Commands** buttons above System Vitals apply one virtual left, right, up, down, open, or close action. **Home** restores the starting stack. A blocked move leaves all blocks in place and explains why. The preview demo is scripted; the separate live controller accepts manual commands and complete ROM-pattern detections from RetroPie game frames. In game play, the player compares the virtual arrangement with the target shown by the game. The flash stream does not tell the controller which block to move or whether the target was achieved.

Pose Lab and recovery

Pose Lab exercises R.O.B.'s head, body turn, arms, and paired grip without game pieces or commands. Its demo is a short greeting; the controls below the scene let the viewer try poses manually. It is an art and motion sandbox, not a third game. In a live session, unreadable flashes produce no movement. A browser reconnect receives the UNO Q's current snapshot; it never restarts an action. If the Gyromite return path disconnects, virtual Controller 2 buttons release even if the browser still shows a gyro on a pad. The game host receives controller states, not a proof that a level was completed.

CHAPTER 04 / USER GUIDE

R.O.B. Vision Setup

Open `http://arduino.local/dashboard/setup.html`. Console Link lists every saved RetroPie and Batocera console separately, with its address and live status. Setup has four checks:

Install the [current release](#) with one command on the UNO Q and one on RetroPie or Batocera. The console command detects its platform and opens first-time pairing automatically. Batocera 43.1 x86_64 is the validated Batocera target; unsupported boards are reported before installation.

1. **Pair the console.** Keep the console installer running while it displays a six-digit code and SHA-256 fingerprint for five minutes. Enter its hostname, code, and fingerprint on Setup, then select **Pair Console**. The UNO Q verifies the certificate fingerprint before sending a unique console credential over the encrypted connection. The console command starts the receiver and configures Gyromite's virtual Controller 2 when pairing completes. Repeat on another console using **Pair Another Console**. Each appears under **Paired Consoles**; **Check Link** counts recent authenticated polls, and an individual row goes offline after three seconds without one.
2. **Check game frames.** On RetroPie, select `lr-robvision-fceumm` or `lr-robvision-nestopia`. On Batocera, launch an exact registered ROM; its R.O.B. Vision FCEUmm wrapper is selected automatically. Setup should show **GAME FRAMES LINKED**. The wrapper forwards each rendered NES light cell to the local receiver, which sends complete commands to the UNO Q. The plain `lr-fceumm` and `lr-nestopia` entries play without this link.
3. **Check Test mode.** Launch the game through its R.O.B. Vision emulator choice and enter Test mode. Gyromite's sustained green field or Stack-Up's alternating green/white frames make R.O.B.'s red head light blink automatically on Mission and Setup. A separately decoded ready-light command makes it glow steadily for a moment. Exiting Test mode returns the matrix to the game's eyes. Neither signal moves arms or game pieces. **Preview Red Light** only demonstrates the animation. If the light stays off, check **GAME FRAMES LINKED** and confirm the game is in Test mode. In Stack-Up, press Select at the ROBOT BLOCK artwork to show the mode list, choose Test, then press Start.
4. **Try the controls.** For Gyromite, **Lower Red** and **Lower Blue** press matching virtual pads immediately; press again to release or choose **Release Both**. Each hold expires after 60 seconds. For Stack-Up, **Left, Right, Up, Down, Open, Close** send one command to the virtual model. Stack-Up has no Gyromite-style Controller 2 return path. Watch Mission's Live Model and Game Table for movement.

A file:// Setup page is only an offline preview. Use the UNO Q URL for live status. Mission keeps Fast Gates and the Game Table next to Pose Preview and System Vitals.

To forget a console, select **Remove** in its row and then **Confirm Remove**. The UNO Q immediately revokes that console's credential; any other paired console remains connected. If the removed console was playing, its game session is cleared. The removed console retains its old local credential, but it can no longer authenticate; rerun the one console install command with `--pair` at the end to reconnect it. Only one console can be the active game source at a time.

CHAPTER 05 / USER GUIDE

R.O.B. Vision installation guide - 0.1.2

R.O.B. Vision runs on an Arduino UNO Q and connects to either RetroPie or Batocera. Install the UNO Q first, then install each console. The release installer downloads a versioned source package, checks its SHA-256 digest, and runs the platform installer. No Git checkout, ROM transfer, or SSH transfer is required.

Before you begin

- Connect the UNO Q and console to the same trusted network. Sign in to the UNO Q as `arduino`, to RetroPie as `pi`, or to Batocera as `root`.
- Install UNO Q App Lab. Standard RetroPie needs Python 3.7+, `gcc`, `openssl`, `sudo`, `systemd`, `uinput`, and at least one installed NES core (`lr-fceumm` or `lr-nestopia`). The UNO Q needs Python 3.9+. Batocera 43.1 x86_64 is the validated Batocera target; the command checks the board and version before installation.
- Exit any running game and EmulationStation on RetroPie. Restarting the virtual joystick while EmulationStation is open can crash its input manager. Stop a different App Lab app before installing R.O.B. Vision. Supply your own legally obtained Gyromite or Stack-Up ROM with an exact name in `config/games.json`.
- `curl`, `tar`, Python 3, and either `sha256sum` or `shasum` are needed to run the single-command download. The installer reports a missing prerequisite before installing.

Install the UNO Q

Run this one command in the UNO Q terminal as `arduino`:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/download/v0.1.2/install.sh | sh -s -- -uno-q
```

The command downloads and verifies R.O.B. Vision, installs the separate App Lab app, and starts it. On upgrades, App Lab stops and restarts the app while preserving its pairing data. The installation includes the UNO Q matrix sketch; App Lab compiles and uploads that sketch when starting the app. It does not replace the board's bootloader or firmware. Open `http://<your-uno-q-hostname>.local/dashboard/setup.html` in a browser on the same network. `http://arduino.local/dashboard/setup.html` is the address of the project's test UNO Q.

Install a console

Run this one command **on RetroPie or Batocera**, replacing your `-uno-q.local` with your UNO Q's LAN hostname or IP address:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/download/v0.1.2/install.sh | \
sh -s -- -console your-uno-q.local
```

The command detects the console. On RetroPie it uses `sudo` for system changes and requires EmulationStation to be closed. On Batocera it runs under the root login and temporarily suspends and resumes its game menu while the virtual joystick is restarted. It installs the frame wrappers, receiver, game launch integration, and Gyromite's virtual Controller 2. For a first install it opens a five-minute pairing window and prints a six-digit code and SHA-256 certificate fingerprint. Enter the console's hostname, code, and fingerprint under **Console Link** on the UNO Q Setup page. Keep the console command open until it reports **Pairing complete**. Successful pairing starts the receiver and maps RetroPie's Player 2 automatically. No second terminal command is required.

Upgrades and re-pairing

An upgrade retains that console's credential and restarts its receiver. Rerun the same command for a repair or upgrade. If you remove a console in Setup, rerun the console command with `--pair` added at the end to open a new pairing window; this is still one command on the console.

The installer leaves ROM files and unrelated games alone. RetroPie registers `lr-robvision-fceumm` and `lr-robvision-nestopia` for the exact registered ROM names. Batocera selects its R.O.B. Vision FCEUmm wrapper for those names while retaining unrelated core choices. Plain NES cores still play games but do not send commands to R.O.B. Vision.

Check the link and play

On Setup, select **Check Link** after pairing. It should show the console online. Launch Gyromite or Stack-Up using a R.O.B. Vision core; Mission and Setup should show **GAME FRAMES LINKED**. In Gyromite Test mode the matrix Test light blinks automatically. In Game A, test blue and red gates independently. Stack-Up Direct mode supplies Left, Right, Up, Down, Open, and Close. The virtual robot and game pieces appear in Mission on a laptop, phone, or tablet. The frame link needs no camera. Game A blue-gate hold and release passed with the corrected Nestopia wrapper on both tested consoles and with FCEUmm on Batocera.

If a game is already running during installation, the console installer stops before changing its configuration. Exit the game and run the same command again. An unsupported Batocera board or version is reported before any changes. For a failed link, see [Troubleshooting](#) and the built-in Help page.

Supported releases and integrity

Version 0.1.2 supports the same validated platforms as 0.1.0: UNO Q App Lab, standard RetroPie with the `pi` account, and Batocera 43.1 x86_64. Other Batocera boards are detected by the same command and rejected until their core libraries and gameplay pass validation. Release assets include the source package, installer, PDFs, and SHA256SUMS. The installer pins the source package's digest; the checksum file also lets you check downloaded assets independently. The repository contains no ROMs.

For development or repair, the [RetroPie](#) and [Batocera](#) guides document individual components. The normal install path is the one command above for each device, followed by browser pairing.

CHAPTER 06 / USER GUIDE

R.O.B. Vision troubleshooting

If EmulationStation exits with an InputManager joystick assertion, return to its shell, stop the R.O.B. Vision receiver if it is repeatedly failing, and rerun the console installer with EmulationStation closed. The installer now checks module compatibility before restarting the receiver.

For installation failures, check the prerequisite or unsupported-device message from the single release command, then rerun that same command. The archive checksum must pass before the installer changes the device. After removing a console in Setup, rerun its command with `--pair`.

First check which page is open: `http://arduino.local/dashboard/` is live; a `file://` page or a simple static web server shows the independent preview.

Symptom	Check
App does not start, port 8766 in use	Stop VirtualGlove or the separate <code>rob-vision.service</code> ; App Lab and that service cannot both bind the port.
GYROMITE / CONNECTED but no robot movement	Check RETROPIE ONLINE and GAME FRAMES LINKED separately. If the latter is absent, restart the game so RetroPie loads the R.O.B. Vision core. Check Mission's Activity Feed for decoded or blocked actions.
RetroPie is online but no game is selected	A receiver can be online with no active game. Launch a registered ROM; then reload Mission if needed.
Test light never blinks	Enter the game's Test mode and confirm GAME FRAMES LINKED with the correct game selected. The light responds automatically to linked game frames. A red-light preview is only an animation demonstration.
Movement command is missed	On RetroPie, confirm GAME FRAMES LINKED and the correct game is selected. The frame link reads every rendered NES frame; check the Activity Feed for a decoded command or a model move blocked by height, grip, or station.
Red/blue gate does not move	Confirm Gyromite is the active RetroArch game, Setup Check Link is online, and RetroArch port 2 is configured. Inspect the game screen during an isolated color hold.
Fast Gates vanished	They show only in live Gyromite.
Gate remains down	Select Release Both ; a hold also expires at 60 seconds. The receiver releases on game exit or stale network data.
Stack-Up block does not move	Review current station, height, gripper, carried group, and destination capacity; invalid moves leave the stack intact and appear in activity.
Demo does not start	Exit the running game first. Demo mode works when paired and idle; Home resets it.
Browser shows stale scene	Reload for a fresh <code>/api/state</code> snapshot. A local preview cannot show UNO Q state.

On the dedicated RetroPie test setup, `systemctl status rob-vision-controller2.service` checks the root virtual-controller service. Inspect RetroPie's runcommand log and configured ROM basename if launch identity is missing. `/dev/input/js1` is expected for the virtual Controller 2 while the player's controller is generally `/dev/input/js0`. Keep tokens and ROM files out of bug reports.